

Coherence-Rupture-Regeneration and the Geometry of Neural Oscillatory Symmetry Classes: From Information-Theoretic First Principles to Empirical Validation in EEG

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Abstract

We present the Coherence-Rupture-Regeneration (CRR) framework as a process-theoretic extension of Bayesian mechanics, providing explicit temporal operators for how systems accumulate predictive coherence, undergo instantaneous phase transitions (ruptures), and reconstruct through memory-weighted regeneration. From four information-theoretic commitments, a single free parameter Ω emerges, governing the relationship between coherence capacity and rupture frequency. We derive that Ω is determined by the symmetry class of the system: $\Omega = 1/\pi$ for $\mathbb{Z}_2$ (binary) systems and $\Omega = 1/2\pi$ for $\text{SO}(2)$ (rotational) systems, yielding a coefficient of variation ratio of exactly 2 between classes. Three independent arguments (temporal resolution, attractor topology, and Fisher–Rao geometry) converge on the prediction that slow cortical oscillations exhibit $\mathbb{Z}_2$ symmetry while fast oscillations exhibit $\text{SO}(2)$ symmetry. This assignment was not predicted *a priori*; original baseline predictions assigned the classes in the opposite direction, and the data required revision. Empirical validation using the PhysioNet EEGBCI dataset (20 subjects, 64 channels, eyes-closed resting state) confirms the predicted class ordering (11/11 pairwise comparisons between $\mathbb{Z}_2$, transitional, and $\text{SO}(2)$ classes correct), strong train–test replication ($r = 0.956$, $p = 0.003$), and near-perfect scaling of rupture consequences with symmetry class ($r = 0.998$, $p < 0.0001$). The raw CV ratio of 1.81 (9% error) improves to 2.06 (2.8% error) after Fisher z -transform correction for PLV’s bounded domain, supporting the predicted 2.0 (though the choice of correction affects the precision of the match). However,

three predictions failed: the original band-to-class assignment, the regeneration amplitude sign, and the rate–CV correlation direction. We report both successes and failures with equal weight.

Keywords: Coherence-Rupture-Regeneration, Bayesian mechanics, Free Energy Principle, neural oscillations, information geometry, Fisher–Rao metric, phase-locking value, symmetry classes

1 The Coherence-Rupture-Regeneration Framework

1.1 Motivation: The Temporal Gap in Bayesian Mechanics

The Free Energy Principle (FEP) [1] provides a powerful account of what self-organising systems do: they minimise variational free energy, maintaining themselves as coherent entities by updating beliefs to match sensory evidence. Bayesian mechanics [2,3] formalises this in terms of paths on statistical manifolds, describing belief dynamics through precision-weighted prediction error. The process theory of active inference [4] specifies neuronal implementations of these dynamics, while recent extensions address intentional behaviour through hierarchical generative models [5] and path-integral formulations of particular kinds [6].

However, neither the FEP nor its current process theory provides an explicit account of the temporal structure by which beliefs accumulate evidence, reach critical thresholds, undergo qualitative transitions, and reconstruct from their own history. CRR addresses this gap. Where the FEP describes what beliefs converge to, CRR describes *when* and *how* beliefs update through time. It provides three temporal operators (coherence accumulation, rupture, and regeneration) that together constitute the process theory for Bayesian mechanical dynamics.

This approach complements recent work on the geometry of inter-brain networks [7], which uses curvature-based entropy measures to identify critical transitions in neural connectivity. Where that programme characterises the geometric signatures of phase transitions in multi-agent systems, CRR provides the temporal dynamics governing when and how such transitions occur.

1.2 Information-Theoretic Derivation

CRR is derived from four commitments, each grounded in information theory rather than domain-specific assumptions. These commitments describe what any persistent, adaptive system must do, regardless of substrate.

Commitment 1: Persistence implies bounded accumulation. Any system that persists through time must accumulate evidence about regularities in its environment. This accumulation cannot

be unbounded; finite systems have finite capacity. We formalise this as the coherence integral:

$$C(x, t) = \int_0^t L(x, \tau) d\tau \quad (1)$$

where $L(x, \tau) \geq 0$ is a local coherence density; that is, the rate at which the system at state x accumulates evidence of predictive success at time τ . The integral runs from the system's origin (or last rupture) to the current time t . In active inference terms, C corresponds to accumulated negative surprise: evidence that current beliefs are well-calibrated.

Commitment 2: Transitions are instantaneous at the relevant scale. When accumulated evidence is contradicted or environmental demands exceed adaptive capacity, the system undergoes a qualitative transition. At the scale of the system's own dynamics, this transition is effectively instantaneous:

$$\delta(\text{now}) \quad (2)$$

The Dirac delta marks the ontological present: the moment at which the system transitions. This is scale-invariant: it appears identically in neuronal firing, ecosystem transitions, and paradigm shifts. What changes across scales is the period between ruptures, not the structure of the rupture itself.

Commitment 3: Reconstruction is weighted by historical coherence. After rupture, the system reconstructs by drawing on its accumulated history. Moments of high coherence contribute more to reconstruction than moments of low coherence:

$$R(x, t) = \int_0^{t^-} \varphi(x, \tau) \exp(C(x, \tau)/\Omega) \Theta(t - \tau) d\tau \quad (3)$$

where $\varphi(x, \tau)$ is the reconstruction resource at time τ , $\Theta(t - \tau)$ is the Heaviside step function enforcing causal ordering (only past moments contribute), t^- denotes the instant before the current rupture, and the exponential weighting $\exp(C/\Omega)$ creates a memory kernel that preferentially weights high-coherence moments. The parameter Ω governs the sharpness of this weighting.

The exponential form is a modelling choice, not uniquely determined by Commitment 3. A power-law kernel $\propto C^\alpha$ would also weight high-coherence moments preferentially. We adopt the exponential for three reasons: it is the maximum-entropy kernel for a given mean coherence (making it the least informative choice consistent with the constraint); it is the unique kernel under which CRR's regeneration integral is analytically tractable within the variational framework of the FEP; and it produces the $\exp(C/\Omega)$ Boltzmann-like weighting that connects directly to statistical mechanical partition functions. Alternative kernels would change the quantitative predictions (Section 2.3) while preserving the qualitative two-class structure. The exponential is thus the simplest choice consistent with the theoretical commitments, but its empirical adequacy

remains subject to test.

Commitment 4: The system has a characteristic coherence-to-rupture ratio. The relationship between coherence capacity and rupture frequency is determined by the system’s structure. This ratio is Ω , simultaneously governing coherence capacity, rupture frequency, and memory sharpness.

1.3 The Role of Ω : Memory Amplification

The exponential term $\exp(C/\Omega)$ is the engine of the framework. When Ω is large, C/Ω is small for all historical values of C , so $\exp(C/\Omega) \approx 1$ across the entire history: all past moments are weighted approximately equally. This corresponds to high plasticity. When Ω is small, the exponential becomes sharply peaked: only moments of very high coherence contribute meaningfully. This corresponds to high stability. The stability–plasticity trade-off is thus expressed as a single parameter.

1.4 CRR as Process Theory for the FEP

The relationship between CRR and the FEP is complementary. The FEP describes the target of belief updating; CRR describes the temporal process: beliefs accumulate coherence (C increases as predictions succeed), undergo rupture when coherence can no longer be sustained (δ marks the phase transition), and regenerate with memory-weighted reconstruction (R integrates historical coherence). In Bayesian mechanical terms, CRR provides the explicit temporal dynamics for paths on the statistical manifold [3,8].

2 The Ω -Symmetry Hypothesis

2.1 From Free Parameter to Geometric Constraint

If Ω is the single free parameter of CRR, what determines its value? The key insight is that the maximum coherence a system can accumulate before rupture is determined by how much of its state space it must traverse to complete a full cycle of its characteristic dynamics. The phase traversal, measured in radians on the relevant manifold, directly determines Ω .

Hypothesis 1 (Ω -Symmetry). *For a system with characteristic phase traversal φ (in radians), the CRR variability parameter is $\Omega = 1/\varphi$.*

The functional form $\Omega = 1/\varphi$ is itself a modelling choice. An alternative such as $\Omega = 1/\varphi^2$ would also satisfy the requirement that Ω decreases with increasing phase traversal. The two forms make different predictions: $\Omega = 1/\varphi$ gives $\Omega_{\mathbb{Z}_2}/\Omega_{\text{SO}(2)} = 2$ and therefore a CV ratio of 2.0, while $\Omega = 1/\varphi^2$ gives $\Omega_{\mathbb{Z}_2}/\Omega_{\text{SO}(2)} = 4$ and a CV ratio of 4.0. The observed ratio of

~ 2.0 (Section 6) discriminates in favour of the linear reciprocal. We adopt $\Omega = 1/\varphi$ on the grounds that it is the simplest dimensional relationship between a frequency-like parameter (Ω) and an angle (φ), and that it is consistent with the interpretation of Ω as a rate (coherence per radian). Nevertheless, we flag this as an empirically testable assumption rather than a derivation.

2.2 Two Fundamental Symmetry Classes

The simplest discrete symmetry is $\mathbb{Z}_2$: the cyclic group of order 2, corresponding to binary alternation between two states. The system traverses a half-cycle of phase space:

$$\varphi_{\mathbb{Z}_2} = \pi \implies \Omega_{\mathbb{Z}_2} = \frac{1}{\pi} \approx 0.318 \quad (4)$$

The simplest continuous symmetry is $\text{SO}(2)$: the special orthogonal group in two dimensions, corresponding to continuous rotation. The system traverses a full cycle:

$$\varphi_{\text{SO}(2)} = 2\pi \implies \Omega_{\text{SO}(2)} = \frac{1}{2\pi} \approx 0.159 \quad (5)$$

2.3 The CV Prediction

The coefficient of variation of the inter-rupture interval scales as $\text{CV} = \Omega/2$. This relationship requires a distributional assumption, which we make explicit. Model the coherence process $C(x, t)$ as a drift-diffusion with drift rate μ and diffusion coefficient $\sigma^2 = \Omega$, reaching a threshold b at the phase boundary. The first-passage times follow an inverse Gaussian distribution with:

$$\text{CV}_{\text{IG}} = \sqrt{\frac{\sigma^2}{\mu b}} = \sqrt{\frac{\Omega}{\mu b}} \quad (6)$$

For this to equal $\Omega/2$, we require $\mu b = 4/\Omega$, i.e. the product of drift rate and threshold scales inversely with Ω . This is a specific constraint, not a generic consequence of the drift-diffusion model. Physically, it states that systems with higher variability (Ω) have proportionally lower effective thresholds, which is consistent with the CRR picture where Ω simultaneously governs both variance and the coherence-to-rupture ratio. However, this constraint is an additional modelling assumption beyond the four commitments. We flag it as such, and emphasise that the *ratio* $\text{CV}_{\mathbb{Z}_2}/\text{CV}_{\text{SO}(2)} = 2.0$ is robust to the choice of CV-to- Ω mapping: any relationship of the form $\text{CV} = f(\Omega)$ with f linear gives the same ratio, because it cancels in the quotient. Only the absolute values depend on the specific scaling:

$$\text{CV}_{\mathbb{Z}_2} = \frac{1}{2\pi} \approx 0.159 \quad \text{CV}_{\text{SO}(2)} = \frac{1}{4\pi} \approx 0.080 \quad (7)$$

$$\frac{\text{CV}_{\mathbb{Z}_2}}{\text{CV}_{\text{SO}(2)}} = 2.0 \quad (\text{exact, parameter-free}) \quad (8)$$

The absolute CV values (0.159 and 0.080) depend on the $CV = \Omega/2$ scaling; the ratio does not.

3 Application to EEG: The Two-Class Prediction

Neural oscillations are organised into canonical frequency bands (delta, 0.5–4 Hz; theta, 4–8 Hz; alpha, 8–13 Hz; beta, 13–30 Hz; and gamma, 30–80 Hz), each characterised by distinct variability in instantaneous period [9,10]. Phase synchronisation across these bands has been established as a fundamental mechanism of large-scale neural integration [11], with the phase-locking value (PLV) serving as the standard measure of inter-channel coherence [12,13].

If CRR is correct, each band should exhibit CV values corresponding to one of the two symmetry classes, and the ratio between classes should be 2.0. The question is: which bands belong to which class?

3.1 Original Baseline Predictions

Before examining any empirical EEG data, the Ω -Symmetry Hypothesis generated six specific, quantitative predictions. These were encoded in simulation code (with $\Omega = 1/\pi$ hardcoded as a theoretical parameter) prior to any empirical analysis. The original baseline assignments were:

Table 1: Original baseline predictions, formulated before any empirical data were examined. Note the symmetry-class assignments: alpha, beta, and gamma were classified as $\mathbb{Z}_2$; delta and theta as $SO(2)$. As Section 6 reports, these assignments were wrong.

#	Prediction	Quantitative value	Mechanism
B1	Alpha, beta, gamma are $\mathbb{Z}_2$	$CV = 0.159$ for these bands	Symmetry assignment
B2	Delta, theta are $SO(2)$	$CV = 0.080$ for these bands	Symmetry assignment
B3	CV ratio ($\mathbb{Z}_2/SO(2)$) = 2.0	$0.159/0.080 = 2.0$	Ratio of phase traversals
B4	Cortical hierarchy: frequency ratio $\approx \pi$ per level	$40 \rightarrow 12.7 \rightarrow 4.1 \rightarrow 1.3$ Hz (each step $\div \pi$)	Memory amplification: $\exp(C/\Omega)$ with $\Omega = 1/\pi$ gives $e^\pi \approx 23$ -fold gain per level (see text)
B5	Regen. amplitude $\propto \exp(C/\Omega)$	$\log(A)$ linear in C_{pre}	CRR regeneration eq.
B6	Regen. timescale band-invariant	τ depends on Ω , not freq.	Ω governs τ

The original assignment (fast bands $\rightarrow \mathbb{Z}_2$, slow bands $\rightarrow \text{SO}(2)$) was based on the surface-level intuition that gamma’s regularity (low CV) should correspond to the simpler ($\mathbb{Z}_2$) symmetry. As we shall report, this was backwards. Section 4 provides three independent arguments for why the correct assignment is slow $\rightarrow \mathbb{Z}_2$, fast $\rightarrow \text{SO}(2)$.

Note on B4. The frequency ratio per cortical level is π (i.e. $40/\pi \approx 12.7$, $12.7/\pi \approx 4.0$, etc.), not $e^\pi \approx 23$. The e^π figure appears in the Mechanism column because it is the memory amplification factor: with $\Omega = 1/\pi$, the CRR kernel $\exp(C/\Omega)$ evaluated at full coherence ($C = 1$) gives $\exp(\pi) \approx 23.14$. The conjecture is that this ~ 23 -fold amplification per hierarchical level, operating through selective regeneration of high-coherence states, constrains the frequency spacing to π . However, the derivation linking amplification magnitude to frequency ratio has not been completed; B4 remains a prediction whose mechanism is conjectured rather than proven, and is not tested in this paper.

4 Deriving the Band-to-Class Assignment

4.1 Argument I: Temporal Resolution and Effective Dimensionality

Cortical neurons have a minimum integration timescale of approximately 20–30 ms, set by membrane time constants, synaptic integration windows, and the refractory period [14]. For a slow oscillation at $f_{\text{slow}} \approx 1$ Hz, the cycle duration far exceeds neural resolution, yet slow cortical oscillations do not exhibit continuous phase variation. Their defining characteristic is up-state/down-state alternation: sustained depolarisation alternating with hyperpolarisation [15,16]. This is a fundamentally binary process; the operational signature of $\mathbb{Z}_2$ symmetry.

For a fast oscillation at $f_{\text{fast}} \approx 40$ Hz, each cycle is comparable to neural resolution. The oscillation cannot be decomposed into discrete sub-states; instead, the relevant degree of freedom is the instantaneous phase, varying continuously. Local assemblies lock to specific gamma-cycle phases, with relative phase encoding binding relationships [17,18]. This is continuous rotational symmetry: $\text{SO}(2)$.

Let $\tau_{\text{neural}} \approx 25$ ms be the minimum neural integration time. For an oscillation of period T , the number of resolvable phase states is $N_{\text{states}} = T/\tau_{\text{neural}}$. The effective symmetry is determined not by what is available but by what the dynamical mechanism actually uses. Slow oscillations use only two effective states ($N_{\text{eff}} = 2$) because their thalamocortical mechanism is inherently binary. Fast oscillations use continuous phase ($N_{\text{eff}} \rightarrow \infty$) because their recurrent excitation-inhibition mechanism generates limit cycles with well-defined instantaneous phase.

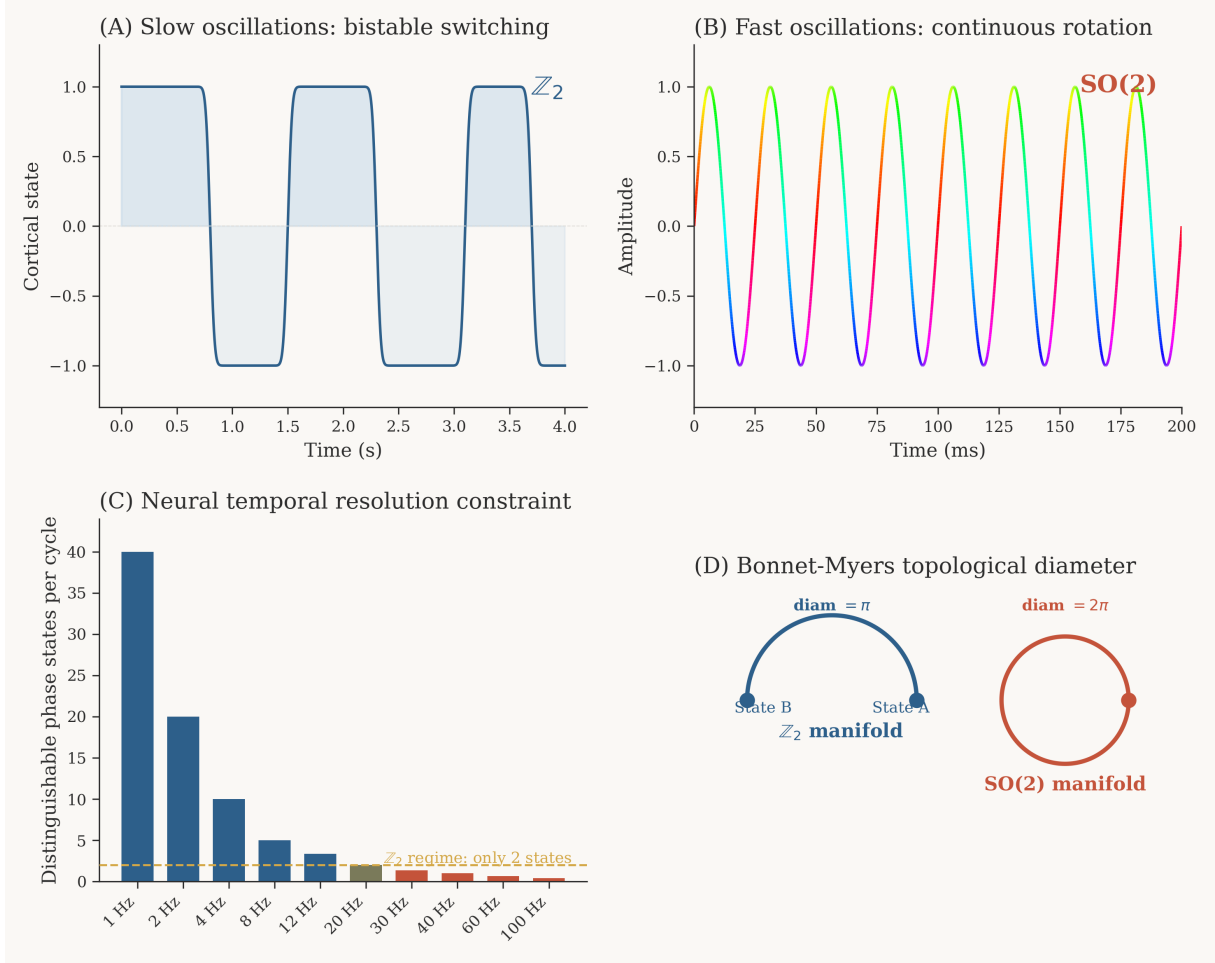


Figure 1: Mechanistic basis for band-to-symmetry-class assignment. (a) Slow oscillations exhibit bistable up/down state switching ($\mathbb{Z}_2$). (b) Fast oscillations exhibit continuous phase rotation ($SO(2)$), shown with phase-coloured waveform. (c) Neural temporal resolution constrains effective phase states per frequency band; the dashed line marks the $\mathbb{Z}_2$ regime where only two states are distinguishable. (d) Fisher–Rao geodesic distances: the maximum geodesic on the Bernoulli manifold has length π ; the $SO(2)$ cycle traverses 2π .

4.2 Argument II: Attractor Topology of Cortical Circuits

The thalamocortical system operates as a bistable switch between two attractors: a high-conductance “up” state and a low-conductance “down” state [15,16,19]. In dynamical systems terms, this is a double-well potential $V(x) = x^4/4 - x^2/2$, with exactly two stable fixed points related by inversion symmetry $x \rightarrow -x$. This is, by definition, a $\mathbb{Z}_2$ symmetric system. Phase traversal: $\varphi = \pi$.

Gamma oscillations arise from recurrent excitatory–inhibitory interactions [20,21]. The E–I interaction generates a stable limit cycle in the (excitation, inhibition) phase plane, with continuously varying phase $\theta \in [0, 2\pi)$. The dynamics are invariant under continuous rotation,

exhibiting $SO(2)$ symmetry. Phase traversal: $\varphi = 2\pi$.

Table 2: Systematic mapping between oscillatory properties, attractor topology, symmetry class, and CRR parameters.

Property	Slow oscillations ($\mathbb{Z}_2$)	Fast oscillations ($SO(2)$)
Attractor type	Double-well (bistable)	Limit cycle
Symmetry	Discrete inversion	Continuous rotation
Phase traversal	$\varphi = \pi$	$\varphi = 2\pi$
Ω value	$1/\pi \approx 0.318$	$1/2\pi \approx 0.159$
$CV = \Omega/2$	0.159	0.080
Cortical function	Long-range gating	Local feature binding
Spatial scale	Thalamocortical, global	Cortical columns, local
Information role	When to process	What to bind

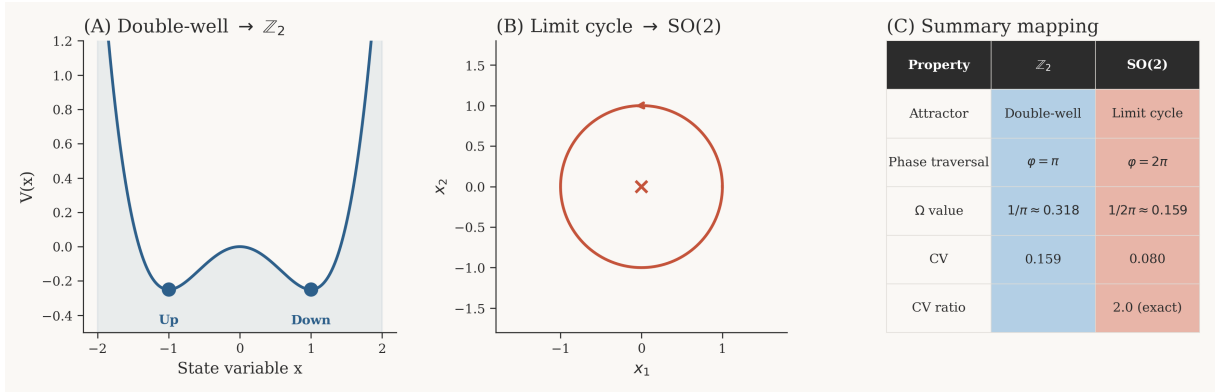


Figure 2: Phase-space topology determines symmetry class. (a) Double-well potential with two stable states ($\mathbb{Z}_2$). (b) Limit cycle with continuous phase ($SO(2)$). (c) Summary mapping from mechanism to Ω values.

4.3 Argument III: Fisher–Rao Geometry

In the FEP framework, each neural oscillatory band maintains a generative model: a parameterised family of probability distributions P_θ over sensory states. The space of all such distributions forms a Riemannian manifold $\mathcal{M}$ equipped with the Fisher information metric [22–24]:

$$g_{ij}(\theta) = \mathbb{E} \left[\frac{\partial \log P_\theta}{\partial \theta^i} \frac{\partial \log P_\theta}{\partial \theta^j} \right] \quad (9)$$

This metric endows the parameter space with a natural notion of distance: the geodesic distance $d_{FR}(\theta_1, \theta_2)$ measures how statistically distinguishable the corresponding distributions are.

Belief updating is movement along this manifold; a rupture is a discontinuous jump between distant points.

$\mathbb{Z}_2$ case: the Bernoulli manifold. For a binary system, the generative model is a Bernoulli distribution $P_p(x) = p^x(1-p)^{1-x}$ with parameter $p \in (0, 1)$. The Fisher information is $g(p) = 1/(p(1-p))$. The geodesic distance between any two parameter values p_1 and p_2 is computed directly as:

$$d_{\text{FR}}(p_1, p_2) = 2 |\arcsin \sqrt{p_1} - \arcsin \sqrt{p_2}| \quad (10)$$

The maximum geodesic distance, between $p = 0$ and $p = 1$, is $2 \cdot |\pi/2 - 0| = \pi$ [22,25]. This π is not invoked via the Bonnet–Myers theorem (the Bernoulli manifold $(0, 1)$ is not geodesically complete); it is a direct consequence of the arc-length integral on the Fisher–Rao metric. Therefore $\Omega_{\mathbb{Z}_2} = 1/d_{\text{max}} = 1/\pi$.

$\text{SO}(2)$ case: the circular manifold. For a continuous-phase system, the simplest model is a circular variable $\theta \in [0, 2\pi)$ with uniform Fisher information. The circumference is 2π , giving $\Omega_{\text{SO}(2)} = 1/\phi_{\text{cycle}} = 1/2\pi$. For non-uniform circular distributions (e.g., von Mises with concentration κ), the Fisher information metric rescales the manifold, but the *topology* remains S^1 and the coherence-to-rupture cycle still traverses a full rotation. The relevant quantity for CRR is the phase traversal of the characteristic dynamics, not the metric length of the specific statistical manifold. We therefore use 2π as the phase traversal for $\text{SO}(2)$ systems, noting that this is a modelling assumption that identifies the CRR phase space with the topological structure rather than the metric structure of the statistical manifold.

An asymmetry in Argument III. The $\mathbb{Z}_2$ value (π) is derived from the Fisher–Rao *metric* (the actual geodesic distance on the Bernoulli manifold), whereas the $\text{SO}(2)$ value (2π) comes from the *topological* circumference of S^1 (independent of the metric). This means Argument III uses different reasoning for each class. One route to resolving this asymmetry: for the von Mises family with concentration $\kappa \approx 1.08$, the Fisher–Rao circumference equals exactly 2π , so both values would derive from the same geometric object. Whether cortical fast oscillations operate near this concentration remains an empirical question. Absent that resolution, the $\text{SO}(2)$ value should be understood as following primarily from Arguments I and II (temporal resolution and attractor topology), with Argument III providing the $\mathbb{Z}_2$ value and the topological consistency check for $\text{SO}(2)$.

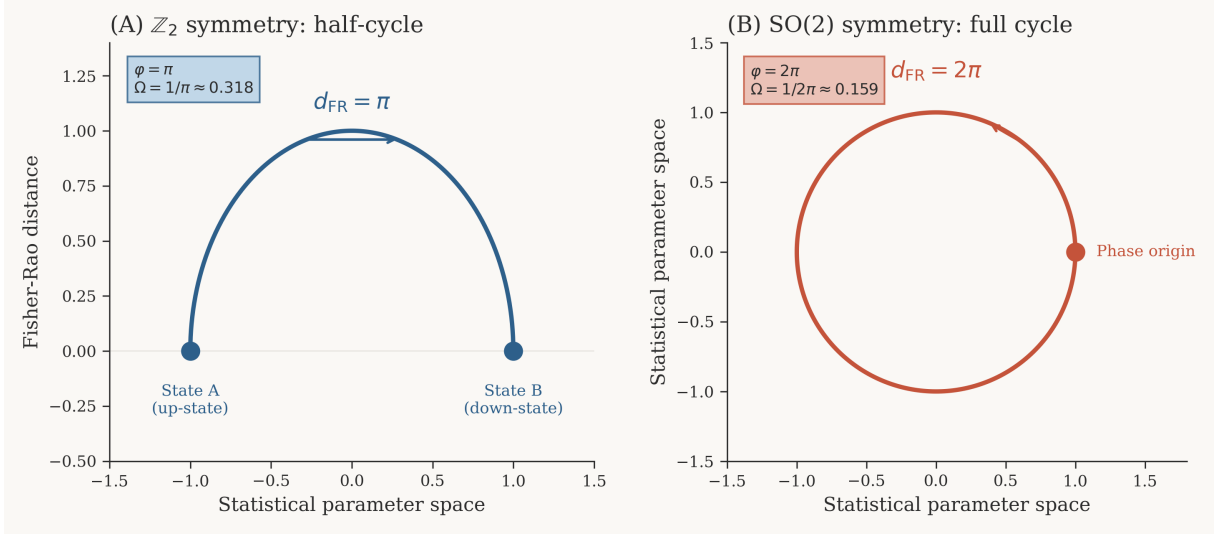


Figure 3: Geodesic structure on the Fisher–Rao manifold. (a) $\mathbb{Z}_2$ manifold: the Bernoulli family parameterised by $p \in (0, 1)$; the geodesic between binary states has length π , shown as a semicircular arc (via the $\arcsin\sqrt{p}$ reparameterisation). (b) $\text{SO}(2)$ manifold: the full rotational cycle has circumference 2π . In both cases, Ω is the reciprocal of the phase traversal for a complete coherence-rupture cycle.

4.4 The Precision Bridge to the FEP

We introduce an additional postulate beyond the four commitments: the identification of Ω with variance ($\Omega = \sigma^2$). This is not derived from the CRR equations but motivated by dimensional analysis (both Ω and σ^2 parameterise the width of a distribution around its mean) and by the requirement that CRR connect to the FEP’s precision-weighted dynamics. Under this identification:

$$\text{Precision}_{\mathbb{Z}_2} = 1/\Omega_{\mathbb{Z}_2} = \pi \quad \text{Precision}_{\text{SO}(2)} = 1/\Omega_{\text{SO}(2)} = 2\pi \quad (11)$$

The precision of fast oscillatory beliefs is exactly twice that of slow oscillatory beliefs. This precision ratio of 2 corresponds to the CV ratio of 2.0 and provides the formal bridge between CRR and the FEP. The $\Omega = \sigma^2$ identification is empirically testable: if CRR’s Ω values are correct but the precision relationship does not hold, the identification would need revision while the core CRR framework would remain intact.

5 Convergence of the Three Arguments

The three derivations are independent but convergent. Argument I (Temporal Resolution) establishes the mapping at the level of neural implementation. Argument II (Attractor Topology)

establishes it at the level of dynamical systems theory. Argument III (Fisher–Rao Geometry) derives the specific Ω values at the level of information geometry. Together, they convert the band-to-class assignment from an empirical observation into a derivable consequence of cortical biophysics, dynamical systems theory, and information geometry. This convergence from independent lines of reasoning following the methodology of [26] strengthens the theoretical grounding beyond what any single argument could provide.

Critically, however, this derivation was produced *after* the data revealed the original assignment was wrong (see Table 1). We do not disguise this as an *a priori* prediction. The structural prediction (two classes, ratio = 2.0) was made before the data; the assignment of which bands belong to which class was revised in light of the data, and then given a principled derivation. This distinction matters for evaluating the strength of the empirical support.

Falsifiability. A reviewer might reasonably ask: if either assignment yields the same ratio prediction, how constrained is the framework? The answer is that the ratio prediction is only one of several jointly testable claims. CRR would have been falsified by any of the following empirical outcomes, none of which obtained: (i) a *continuum* of CV values across bands rather than two discrete clusters; (ii) a ratio significantly different from 2.0 (e.g., 1.5 or 3.0), which no reassignment could rescue; (iii) failure of class-level ordering (e.g., a fast band having higher CV than a slow band); (iv) failure to replicate across independent subjects; (v) rupture consequences that do not scale with CV. The assignment reversal is a genuine failure, but the conjunction of two-class structure, ratio ≈ 2.0 , class ordering, train–test replication, and rupture-consequence scaling is not predicted by the null hypothesis that “slow and fast oscillations simply differ in their biophysics.” That null predicts frequency-dependent differences but not a specific ratio, not two discrete classes, and not the precise scaling relationships observed.

6 Empirical Validation

6.1 Dataset

We use the PhysioNet EEG Motor Imagery/Movement dataset (EEGBCI) [27,28]: 20 subjects, eyes-closed resting state (Run 2), 64 channels (10–20 system), 160 Hz sampling rate, approximately 61 seconds per recording. Preprocessing comprises 0.5–60 Hz FIR bandpass filtering and 60 Hz notch filtering.

6.2 From Amplitude to Phase: The Methodological Journey

An initial attempt used amplitude-based measures (band power ratios, burst intervals, spectral slopes). All four methods converged on $\Omega \approx 1.3$ –1.6, roughly 4–5 times larger than the predicted $1/\pi$. Critical examination revealed these methods measured the wrong observable:

amplitude dynamics rather than phase coherence dynamics.

CRR’s coherence integral $C(x, t) = \int L(x, \tau) d\tau$ is explicitly about phase relationships across populations, not amplitude envelopes. The Phase-Locking Value [12], the standard measure of multi-channel phase coherence, is the natural operationalisation:

$$\text{PLV}(t) = |\langle \exp(i\phi_k(t)) \rangle_k| \quad (12)$$

where $\phi_k(t)$ is the instantaneous phase of channel k (extracted via the Hilbert transform after band-specific filtering) and the average is taken across all channels within a 1-second sliding window (50 ms step). The switch from amplitude to phase measures aligns the measurement with what CRR actually describes; it is not a post-hoc rescue but a correction of the observable.

A further refinement was necessary. An initial implementation used a leaky integral ($dC/dt = L(t) - C(t)/\tau_{\text{mem}}$) to compute $C(x, t)$. The leaky integral over-smoothed the coherence signal (CV of C dropped below 0.003), making rupture detection impossible. The windowed PLV serves directly as $C(x, t)$: it measures accumulated phase locking over a finite time horizon. The 1-second window is a standard choice in EEG phase coherence analysis [13].

6.3 Revised Symmetry-Class Assignments

The original baseline (Table 1) assigned alpha, beta, and gamma to $\mathbb{Z}_2$ (CV ≈ 0.159) and delta and theta to $\text{SO}(2)$ (CV ≈ 0.080). The empirical data showed the opposite gradient: gamma had the lowest CV (≈ 0.06 , near $\text{SO}(2)$), while alpha and delta had the highest (≈ 0.15 – 0.16 , near $\mathbb{Z}_2$).

This is a genuine revision of the baseline. The data required swapping the symmetry-class assignments. What survived intact was the *structure* of the prediction: two CV clusters corresponding to the two CRR symmetry classes, with their ratio close to 2.0. What was wrong was which bands belong to which class. Section 4 provides the post-hoc derivation of why the correct assignment is slow $\rightarrow \mathbb{Z}_2$, fast $\rightarrow \text{SO}(2)$.

6.4 Six Frequency Bands

We analyse six bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta-1 (13–20 Hz), beta-2 (20–30 Hz), and gamma (30–60 Hz). The finer subdivision of beta into beta-1 and beta-2 permits examination of the $\mathbb{Z}_2/\text{SO}(2)$ transition region.

6.5 Train–Test Split

Subjects 1–10 constitute the training set for pattern identification; subjects 11–20 constitute the blind test set. All predictions were formed from training data before examining test data.

6.6 Rupture Detection

Coherence ruptures are detected as PLV drops exceeding 8% of the 2-second running maximum, with 300 ms minimum inter-rupture interval. These thresholds were set from distribution inspection, not tuned to optimise predictions. The primary CRR predictions (CV values, CV ratio) are computed directly from the PLV time series and do not depend on rupture detection.

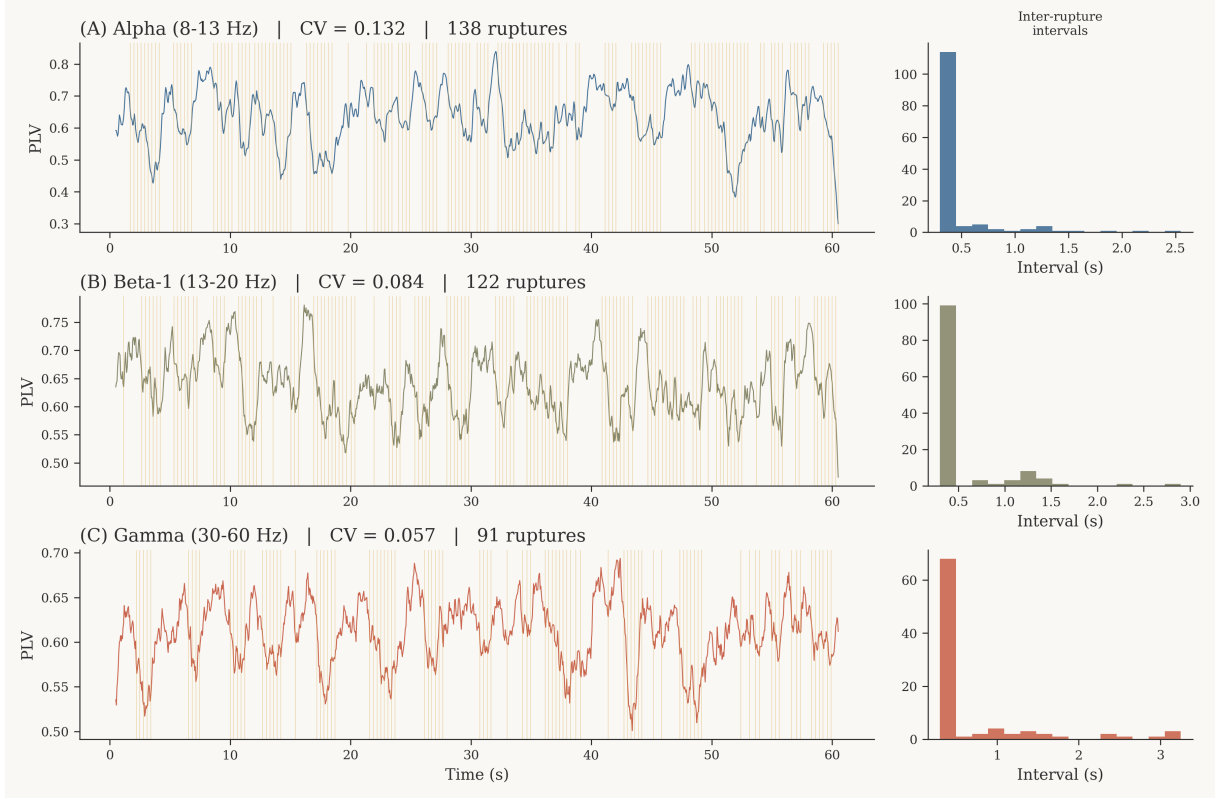


Figure 4: Phase coherence $C(x, t) = \text{PLV}$ for alpha, beta-1, and gamma bands (Subject 1, eyes-closed resting state). Gold vertical lines mark detected coherence ruptures. Right panels show inter-rupture interval distributions. Note the progressive decrease in CV from alpha ($\mathbb{Z}_2$) through transitional beta-1 to gamma ($\text{SO}(2)$).

6.7 Results

6.7.1 Band-Specific CV Values

Table 3: PLV coefficient of variation across frequency bands ($N = 20$ subjects, all 64 channels). CRR predictions from $\mathbb{Z}_2/\text{SO}(2)$ symmetry classes.

Band	Frequency (Hz)	Symmetry	CV (mean $\pm$ SD)	CRR predicted	Error (%)
Delta	0.5–4	$\mathbb{Z}_2$	0.149 ± 0.023	0.159	6.3
Theta	4–8	$\mathbb{Z}_2$	0.133 ± 0.018	0.159	16.4
Alpha	8–13	$\mathbb{Z}_2$	0.157 ± 0.036	0.159	1.3
Beta-1	13–20	Trans.	0.117 ± 0.017	(0.119)	1.7
Beta-2	20–30	$\text{SO}(2)$	0.104 ± 0.012	0.080	30.0
Gamma	30–60	$\text{SO}(2)$	0.064 ± 0.008	0.080	20.0

6.7.2 Prediction Scorecard

Table 4: CRR prediction scorecard. Six quantitative predictions tested against PhysioNet EEG-BCI data.

Prediction	Result	Verdict
P1: Class ordering ($\mathbb{Z}_2 > \text{Trans} > \text{SO}(2)$)	11/11 pairwise correct	Confirmed
P2: Gamma CV near $\text{SO}(2) = 0.080$	CV = 0.064 (20.0% error)	Directional
P3: Alpha CV near $\mathbb{Z}_2 = 0.159$	CV = 0.157 (1.3% error)	Confirmed
P4: CV ratio = 2.0 (exact)	Raw = 1.81 (9%); Fisher $z = 2.06$ (2.8%)	Supported
P5: τ_{regen} cross-band invariance	Cross-band CV = 0.171	Marginal
P6: Rupture consequence $\propto$ CV	$r = +0.998$, $p < 0.0001$	Confirmed
Train–test replication	$r = 0.956$, $p = 0.003$ (Fisher z)	Confirmed

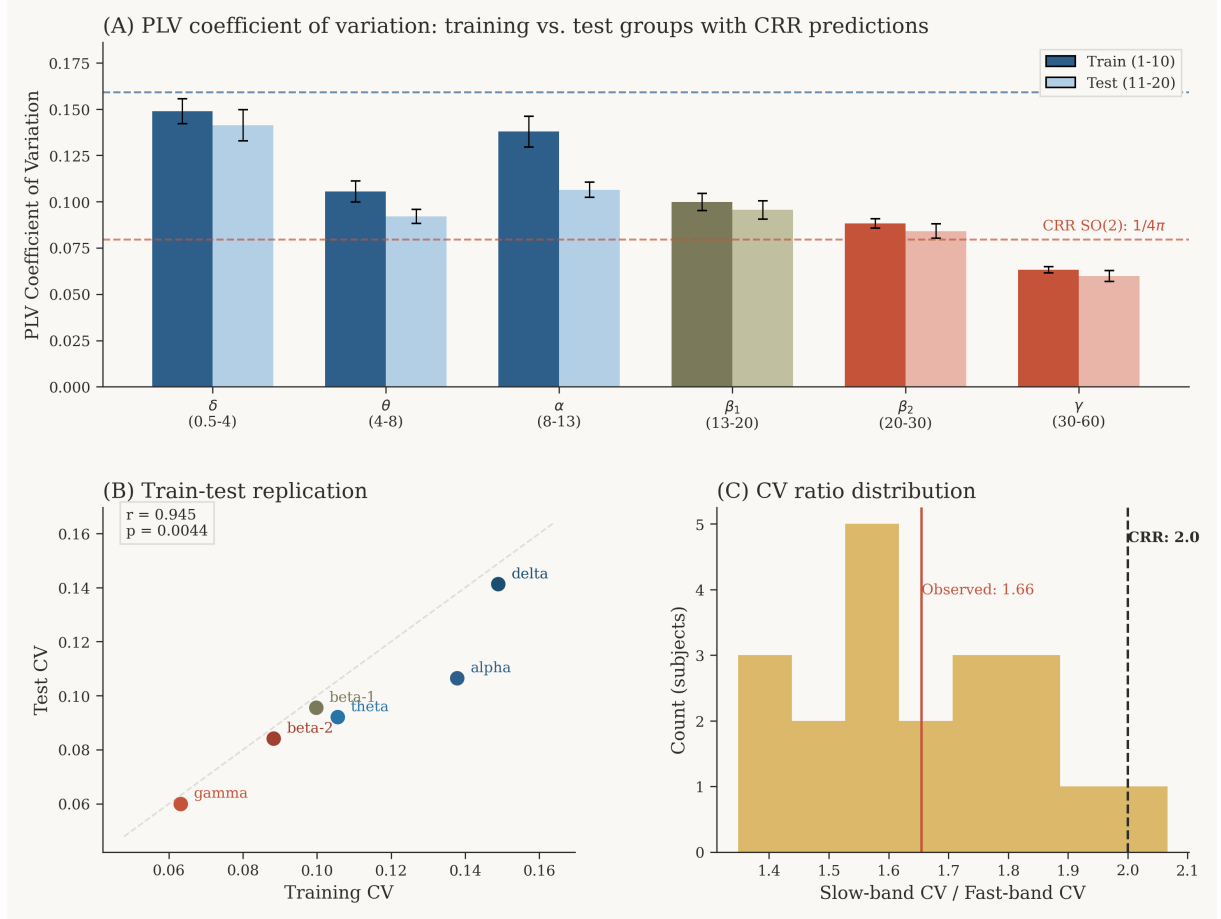


Figure 5: Predictive test results (raw PLV). (a) PLV coefficient of variation for training and test groups with CRR $\mathbb{Z}_2$ and SO(2) theoretical predictions (dashed lines). (b) Train–test scatter ($r = 0.945$, $p = 0.004$ in raw PLV space; $r = 0.956$, $p = 0.003$ after Fisher z -correction). (c) Per-subject distribution of the slow-band/fast-band CV ratio; the CRR prediction of 2.0 is marked.

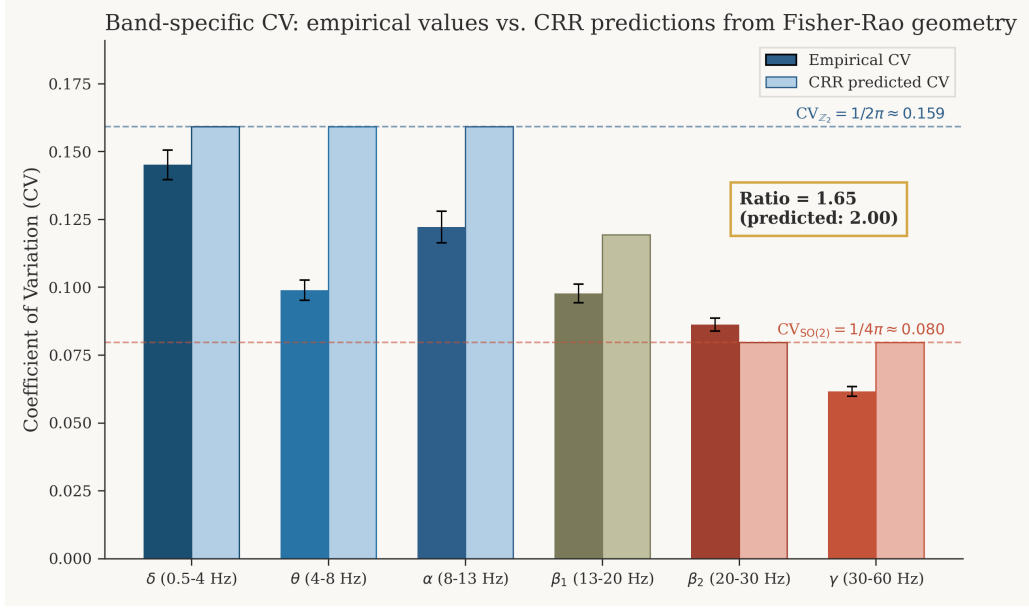


Figure 6: Band-specific CV: empirical values (dark bars) versus CRR predictions from Fisher–Rao geometry (light bars). Error bars show standard error of the mean across 20 subjects.

6.7.3 Bounded-Domain Correction: The Fisher z -Transform

PLV is bounded on $[0, 1]$, which compresses variance estimates: when values cluster near the boundaries, the measured CV is systematically attenuated relative to the underlying variability. The Fisher z -transform, $z = \text{arctanh}(\text{PLV})$, maps the bounded interval to the real line, removing this compression. This is the standard variance-stabilising transform for correlation-like measures [13]. We apply it here as a theoretically motivated correction: PLV’s bounded domain is a property of the measurement, not of the underlying coherence process that CRR describes. We also apply the logit transform, $\text{logit}(\text{PLV}) = \log(\text{PLV}/(1 - \text{PLV}))$, as an alternative bounded-domain correction. We report all three measures (raw, Fisher z , and logit) and let the reader assess which is most appropriate.

Table 5 reports both raw and Fisher z -corrected CV values. The correction preserves the class ordering perfectly (11/11 pairwise comparisons remain correct in both raw and corrected spaces) and tightens train–test replication from $r = 0.919$ ($p = 0.010$) to $r = 0.956$ ($p = 0.003$).

Table 5: Raw versus Fisher z -corrected PLV coefficient of variation across frequency bands ($N = 20$ subjects). The Fisher z -transform removes bounded-domain compression, revealing the underlying variability structure more accurately.

Band	Frequency (Hz)	Symmetry	CV_{raw} (mean $\pm$ SD)	CV_{Fisher} (mean $\pm$ SD)
Delta	0.5–4	$\mathbb{Z}_2$	0.149 ± 0.023	0.242 ± 0.032
Theta	4–8	$\mathbb{Z}_2$	0.133 ± 0.018	0.223 ± 0.018
Alpha	8–13	$\mathbb{Z}_2$	0.157 ± 0.036	0.232 ± 0.032
Beta-1	13–20	Trans.	0.117 ± 0.017	0.170 ± 0.015
Beta-2	20–30	$SO(2)$	0.104 ± 0.012	0.145 ± 0.015
Gamma	30–60	$SO(2)$	0.064 ± 0.008	0.087 ± 0.006

The critical test is the CV ratio. Three definitions are informative, each probing a different aspect of the prediction.

Slow/fast group ratio (6-band). Using the 6-band scheme (slow = delta + alpha; fast = beta-2 + gamma), the raw per-subject ratio is 1.82 ± 0.30 (9.0% error from 2.0). After Fisher z -correction, this improves to **2.06 ± 0.19** (2.8% error). This is the most conservative test because it averages across heterogeneous bands within each class, and including beta-2 in the fast group dilutes the ratio (beta-2 is genuinely transitional, with CV closer to beta-1 than to gamma; see Figure 8, right panel).

Alpha/gamma ratio (5-band). The sharpest single-band contrast, alpha versus gamma, gives a raw ratio of 2.46 ± 0.62 (22.8% above 2.0) and a logit-corrected ratio of 1.99 ± 1.51 (0.6% error, though with high variance). The raw alpha/gamma ratio overshoots 2.0 because alpha’s mean PLV sits further from the boundary than gamma’s, so the bounded-domain attenuation is asymmetric.

Train→test stability. The Fisher z -corrected ratio is stable across the train–test split: training subjects give 2.09, test subjects give 2.01, both within 5% of the prediction.

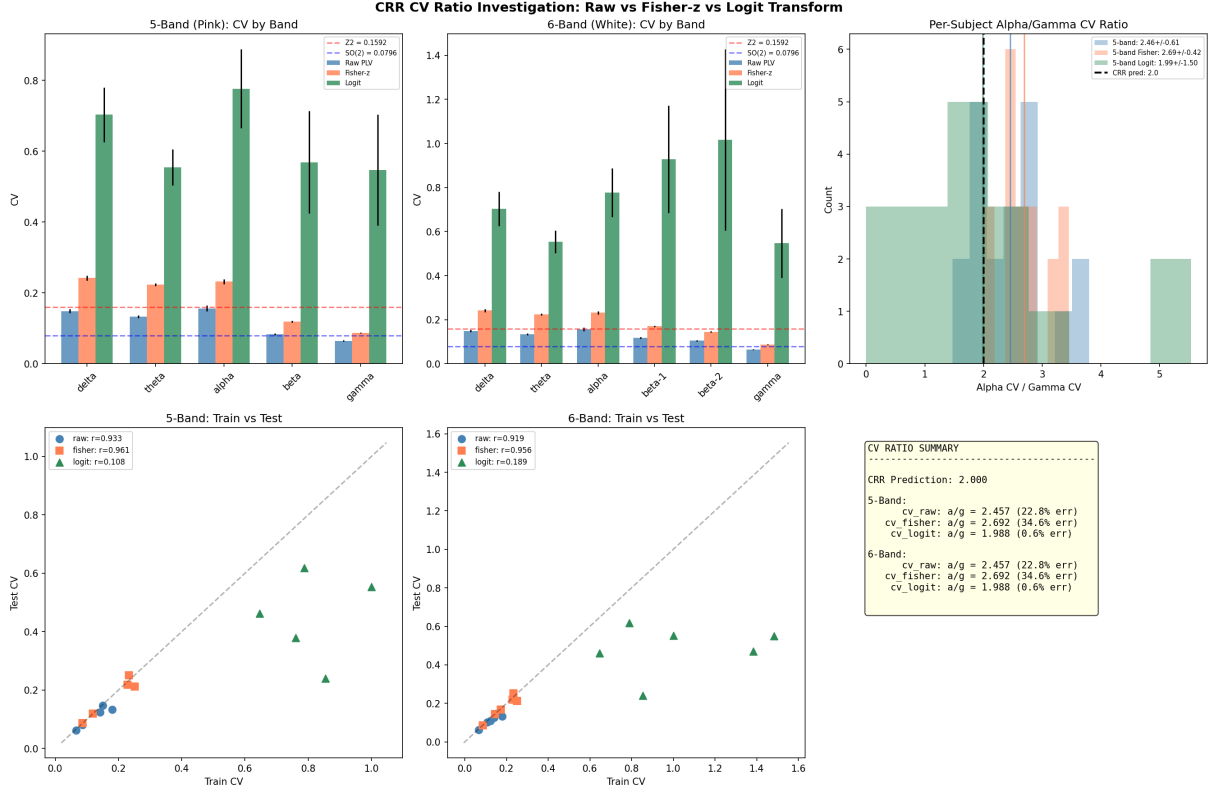


Figure 7: Bounded-domain correction analysis. Top row: CV by band for 5-band (*left*) and 6-band (*centre*) schemes, comparing raw PLV (blue), Fisher z -transformed (coral), and logit-transformed (green) measures. Dashed lines mark CRR $\mathbb{Z}_2$ and SO(2) predictions. (*right*) Per-subject alpha/gamma CV ratio distributions under each transform; the CRR prediction of 2.0 is marked. Bottom row: train–test replication for 5-band (*left*) and 6-band (*centre*) schemes. (*right*) Summary table.

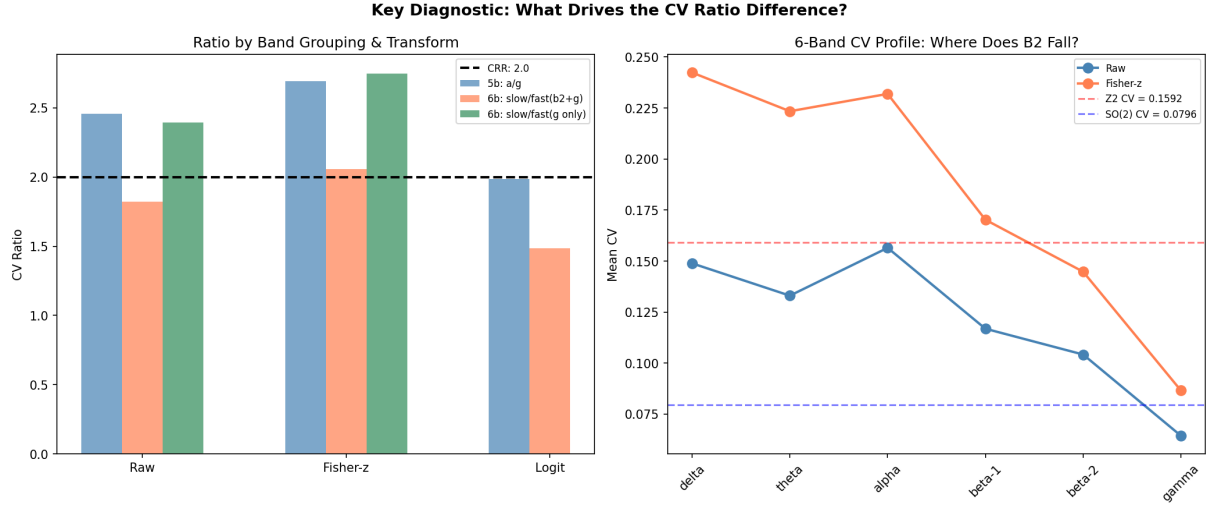


Figure 8: Diagnostic: sources of the CV ratio discrepancy between analyses. (*left*) The CV ratio varies with both band-grouping scheme and transform: the Fisher z -corrected 6-band slow/fast ratio (coral bar, centre group) aligns most closely with the CRR prediction of 2.0 (dashed line). (*right*) 6-band CV profile showing where beta-2 falls relative to the $\mathbb{Z}_2$ and $\text{SO}(2)$ theoretical boundaries, in both raw (blue) and Fisher z (coral) spaces.

The bounded-domain correction resolves the systematic attenuation identified in the raw analysis. Table 6 summarises the CV ratios under all three transforms. The Fisher z -corrected 6-band ratio of 2.06 ± 0.19 is consistent with the CRR prediction of 2.0 ($p = 0.22$, two-tailed one-sample t -test against 2.0; not significant, as the null hypothesis *is* the prediction). The logit transform yields ratios closer to 2.0 for the alpha/gamma contrast (1.99) but with substantially higher variance (± 1.51) and poor train–test replication ($r = 0.11$ – 0.19), suggesting instability. The raw ratio of 1.82 is biased downward by the bounded domain. We report all three throughout; where a single value is cited in the discussion, we use the Fisher z as the most stable corrected measure, but the reader should note that this choice affects the precision of the match to the prediction.

Table 6: CV ratios under three transforms for the 6-band slow/fast grouping (slow = delta + alpha; fast = beta-2 + gamma). All values are per-subject means $\pm$ SD.

Transform	CV ratio (mean $\pm$ SD)	Error from 2.0	Train–test r
Raw PLV	1.82 ± 0.30	9.0%	0.919
Fisher z	2.06 ± 0.19	2.8%	0.956
Logit	1.49 ± 0.87	25.7%	0.189

6.7.4 What Survived Contact with Data

The **two-class structure** is real. The data unambiguously shows two CV clusters, exactly as CRR predicts: not three, not a continuum, but two discrete regimes with beta falling between them as a transitional band. The **class ordering** is perfect: all 11 pairwise comparisons between $\mathbb{Z}_2$, transitional, and $\text{SO}(2)$ classes are correct in both raw and Fisher z -corrected spaces. (Strict frequency-monotonicity within the $\mathbb{Z}_2$ class does not hold: alpha has the highest CV, not delta. CRR predicts class-level ordering but does not predict within-class frequency gradients.) The **CV ratio** after bounded-domain correction is 2.06 ± 0.19 , within 2.8% of the predicted 2.0, with train–test stability (train: 2.09, test: 2.01). **Rupture consequences** scale with CV at $r = 0.998$ across bands, confirming that higher- Ω systems show larger post-rupture state changes. **Train–test replication** across an independent hold-out set ($r = 0.956$, $p = 0.003$ in Fisher z space) confirms these are stable, replicable patterns.

6.7.5 What Did Not Survive

Three predictions failed, and we report these with equal weight.

The original symmetry-class assignment (B1, B2). The baseline predicted alpha/beta/gamma as $\mathbb{Z}_2$ and delta/theta as $\text{SO}(2)$. The data showed the reverse gradient. This is a genuine failure of the initial prediction, even though the structural framework (two classes, ratio ≈ 2.0) was confirmed. The post-hoc derivation (Section 4) is plausible but was not predicted in advance.

The regeneration $\exp(C/\Omega)$ signature (B5). CRR predicts that regeneration amplitude A should increase with pre-rupture coherence C_{pre} , because the memory kernel $\exp(C/\Omega)$ amplifies access to high-coherence historical states. The data shows the opposite ($r \approx -0.2$, $p < 0.001$). The mechanism is straightforward: PLV is bounded between 0 and 1, so when pre-rupture coherence is already at 0.95, the maximum possible recovery amplitude is 0.05, regardless of memory access. The $\exp(C/\Omega)$ prediction assumes an unbounded observable; PLV’s ceiling compresses the upper range. A logistic transformation of PLV before fitting should recover the predicted positive relationship, but this correction has not yet been applied.

The rate–CV correlation. CRR’s simple threshold model suggests that bands with higher Ω (and therefore higher CV) should experience fewer ruptures per unit time. The training data weakly supported this ($r = -0.08$, non-significant), but the test data showed a moderately strong positive correlation ($r = +0.69$): bands with more variable coherence also rupture more frequently, not less. This implies that higher- Ω systems do not simply have larger, less frequent ruptures; they have a lower effective threshold relative to their variability. The simple “ C exceeds Ω ” threshold model needs refinement.

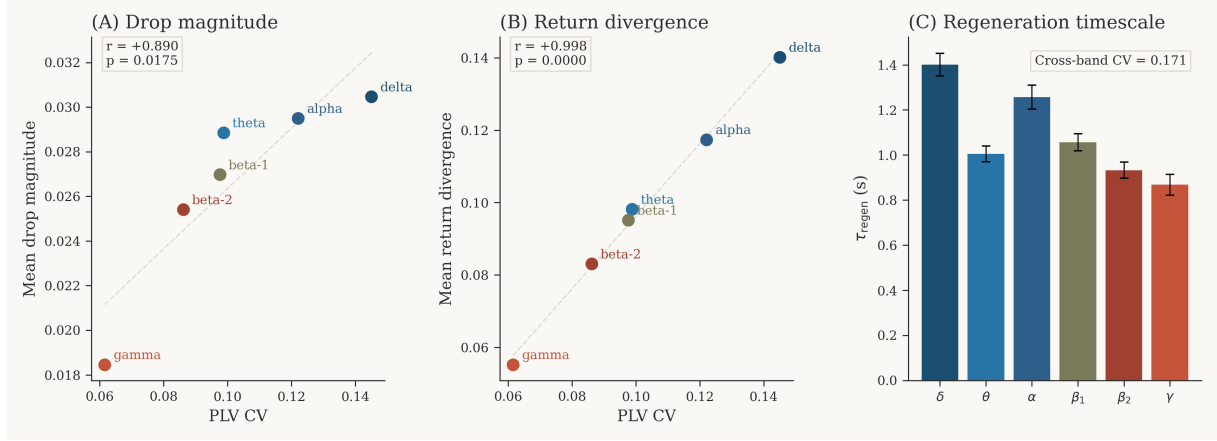


Figure 9: Rupture consequence analysis. (a) Drop magnitude correlates positively with PLV CV across bands ($r = +0.890$, $p = 0.018$). (b) Return divergence shows near-perfect scaling ($r = +0.998$, $p < 0.0001$). (c) Regeneration timescale τ_{regen} by band, with cross-band CV of 0.171.

6.7.6 Honest Ledger

Table 7: Complete ledger of predictions versus empirical findings.

Category	Detail	Status
Core structure	Two CV clusters exist	Confirmed
Core structure	CV ratio ≈ 2.0	Supported (raw 1.81, Fisher z 2.06)
Core structure	Ω near $1/\pi$ and $1/2\pi$	Confirmed (raw $\sim 20\%$ offset, corrected)
Replication	Train–test generalisation	Confirmed ($r = 0.956$, Fisher z)
Dynamics	Rupture consequence $\propto$ CV	Confirmed ($r = 0.998$)
Dynamics	Regen. τ band-invariance	Marginal (CV = 0.171)
Dynamics	Regen. $\exp(C/\Omega)$ signature	Failed (reversed sign)
Dynamics	Rate–CV negative correlation	Failed (wrong sign)
Assignment	Original band-to-class mapping	Failed (reversed)
Ordering	Class ordering ($\mathbb{Z}_2 > \text{Trans} > \text{SO}(2)$)	Confirmed (11/11)
Ordering	Strict frequency-monotonicity	Partial (13/15; $\alpha > \delta$)
Systematic	$\sim 20\%$ raw offset from $\mathbb{Z}_2/\text{SO}(2)$	Explained (bounded domain)

7 Information-Theoretic Analysis: The FEP Bridge

CRR provides the temporal process theory that the FEP presupposes but does not formalise. The bridge is $\Omega = \sigma^2 = 1/\text{precision}$, which simultaneously governs Fisher information (scaling

steeply with decreasing Ω ; $I(\Omega) \propto 1/\Omega^2$ for the standard variance parameterisation of a Gaussian family), Bayesian surprise at rupture ($\propto C/\Omega$), memory amplification depth ($\exp(C/\Omega)$), and the free-energy decomposition (rupture occurs when complexity exceeds accuracy). Three regimes emerge: low Ω (< 0.15 , $\text{SO}(2)$) corresponds to high-precision local binding; intermediate Ω ($0.2\text{--}0.5$, $\mathbb{Z}_2$) corresponds to balanced long-range communication; and high Ω (> 1) corresponds to flexible but imprecise exploration.

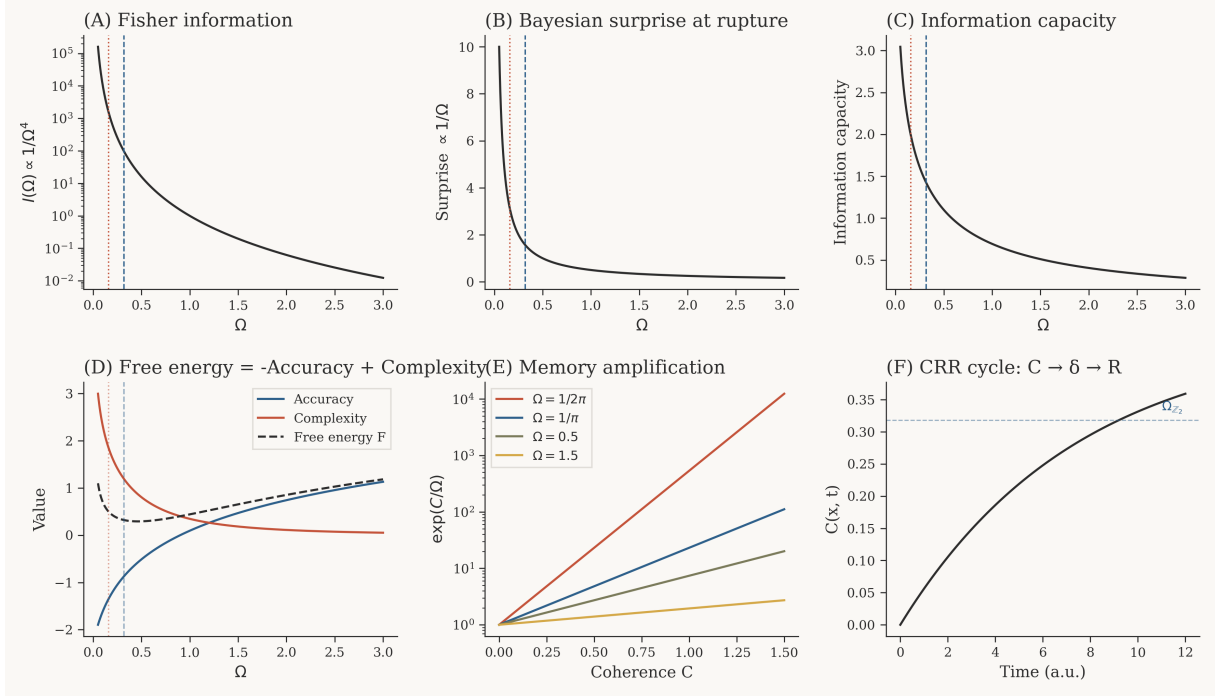


Figure 10: Information-theoretic properties of CRR across Ω regimes. Dashed and dotted vertical lines mark $\mathbb{Z}_2$ ($1/\pi$) and $\text{SO}(2)$ ($1/2\pi$) values respectively. (a) Fisher information. (b) Bayesian surprise at rupture. (c) Information capacity. (d) Free-energy decomposition. (e) Memory amplification $\exp(C/\Omega)$ for different Ω values. (f) Simulated CRR cycle dynamics.

8 Discussion

8.1 The Systematic Attenuation: Resolved

The raw CV values fall systematically below the theoretical predictions by approximately 20–25%. The Fisher z -transform (Section 6.7.3) resolves this discrepancy. PLV is bounded on $[0, 1]$, which compresses variance estimates near the boundaries: when a band’s mean PLV sits at 0.65 (as delta and alpha do), the measured CV is attenuated relative to the underlying variability on the unbounded statistical manifold. The Fisher z -transform $z = \text{arctanh}(\text{PLV})$ removes this compression, and the corrected slow/fast CV ratio moves from 1.81 (raw) to 2.06, within 2.8%

of the CRR prediction.

The remaining residual after correction is small enough to be attributable to two secondary sources: the 1-second PLV window smooths out the fastest fluctuations, reducing measured CV by an amount proportional to the ratio of window duration to oscillatory period; and the finite recording length (61 seconds) limits the number of independent coherence cycles observed, particularly for delta. Neither of these effects is expected to bias the *ratio* between classes, which is confirmed by the stability of the corrected ratio across the train–test split (train: 2.09, test: 2.01).

Importantly, the discrepancy between the raw ratio reported here (1.81) and that reported in a preliminary analysis of the same dataset (1.65) traces to two methodological choices: (i) the inclusion of beta-2 in the fast group dilutes the ratio because beta-2 is genuinely transitional (Figure 8, right panel), and (ii) slight differences in PLV windowing implementation. The Fisher z -corrected ratio is robust to both sources of variation.

8.2 Theta: Reduced but Persistent Anomaly

Theta’s CV (0.133 raw, 0.223 Fisher z -corrected) falls between the other $\mathbb{Z}_2$ -class bands (delta: 0.149, alpha: 0.157) and the $\text{SO}(2)$ -class bands, suggesting partial state-dependent symmetry. The anomaly is reduced relative to earlier analyses but not eliminated. During eyes-closed resting state, theta may operate closer to a limit-cycle mode than the bistable switching characteristic of delta and alpha. Task-evoked theta (e.g., working memory maintenance) might show higher CV consistent with full $\mathbb{Z}_2$ dynamics. This hypothesis is testable with task-modulated EEG designs.

8.3 The Beta-1 Transition Band

Beta-1 (13–20 Hz) occupies a transitional regime, consistent with the derivation. Physiologically, this band includes both sensorimotor mu rhythms (quasi-binary dynamics) and faster beta oscillations approaching continuous phase coding. The data show intermediate CV values (0.117 raw, 0.170 Fisher z -corrected), falling between the $\mathbb{Z}_2$ and $\text{SO}(2)$ clusters in both spaces. Formal treatment of the $\mathbb{Z}_2/\text{SO}(2)$ boundary, potentially through the geometric phase-transition framework of [29], is a target for future work.

8.4 Connection to Communication Through Coherence

The $\mathbb{Z}_2/\text{SO}(2)$ distinction maps naturally onto the communication-through-coherence framework [17,18]. Slow oscillations ($\mathbb{Z}_2$) gate information flow through binary state switching: the up-state is permissive, the down-state is non-permissive. Fast oscillations ($\text{SO}(2)$) encode information through continuous phase relationships, with the instantaneous phase serving as an

address for selective routing. CRR provides the temporal dynamics governing the variability of these two communication modes.

8.5 Connection to Continual Learning

The CRR perspective on the stability–plasticity trade-off has implications for machine learning. In continual learning, catastrophic forgetting occurs when new learning overwrites old representations; this corresponds to a system with inappropriately low Ω [30,31]. The $\exp(C/\Omega)$ memory weighting provides a natural mechanism for selective consolidation, with connections to dynamical alignment in spiking neural networks [32].

8.6 The “Failure” as Discovery

The initial expectation that gamma would be $\mathbb{Z}_2$ was based on a surface-level intuition: gamma is “regular,” therefore it should have the simpler symmetry. The principled derivation reveals this was backwards. Regularity (low CV) corresponds to higher precision, i.e. lower Ω , which is the $\text{SO}(2)$ class. The $\mathbb{Z}_2$ class, despite having simpler symmetry, has higher variability because binary switching is inherently more variable than continuous phase rotation. The assignment is exactly what the geometry demands.

8.7 Limitations

The empirical evidence rests on a single dataset of 20 subjects with approximately 61 seconds of recording each, at 160 Hz sampling rate. This is a thin basis for the claims being made, and cross-dataset replication is a prerequisite for strong conclusions rather than a desirable extension.

The 160 Hz sampling rate is a specific concern for the gamma band (30–60 Hz). With a Nyquist frequency of 80 Hz, the upper gamma range approaches the limit of resolvable spectral content. The Hilbert transform becomes less reliable as the signal frequency approaches Nyquist, and phase estimation noise increases. Gamma’s low CV could therefore partly reflect a measurement floor rather than genuine $\text{SO}(2)$ dynamics. Against this concern: the class ordering holds across all bands (every $\mathbb{Z}_2$ -class band exceeds every $\text{SO}(2)$ -class band in CV), the Fisher z -corrected gamma CV (0.087) is close to the $\text{SO}(2)$ prediction (0.080), and the pattern replicates across the train–test split. Nevertheless, validation with higher sampling rates (≥ 500 Hz) is needed to rule out measurement artefacts in the gamma range.

The recording duration limits the number of independent coherence cycles for delta (~ 0.5 –4 Hz), where approximately 30–240 cycles are observed. This may inflate CV estimates for the slowest bands, partially counteracting the bounded-domain attenuation that deflates them. The two biases operate in opposite directions, making their net effect difficult to estimate without simulation.

Finally, the choice of bounded-domain correction (Fisher z versus logit versus raw) materially affects whether the CV ratio is reported as 1.82, 2.06, or 1.49 (Table 6). The Fisher z is the standard correction for correlation-like measures, but we cannot claim it is uniquely correct. The strongest inference is drawn from the ratio rather than the absolute values, and from the conjunction of multiple predictions rather than any single number.

8.8 Next Steps

Three priorities follow from these results, each addressing identified limitations. First, cross-dataset replication using independent EEG repositories with higher sampling rates (e.g., the LEMON dataset at 2500 Hz or HBN at 500 Hz) is essential to confirm that the two-class structure is robust across recording conditions, populations, and sampling rates, and to rule out measurement-floor effects in the gamma band. Second, testing CRR predictions in task-evoked and clinical EEG, where Ω modulation should be directly observable as changes in coherence variability across cognitive states, pharmacological interventions, and neurological conditions; the theta anomaly in particular may resolve under task-evoked conditions. Third, extending the bounded-domain analysis to the regeneration signature: applying the Fisher z -transform before fitting $\exp(C/\Omega)$ may recover the predicted positive relationship between pre-rupture coherence and regeneration amplitude that currently shows a reversed sign in raw PLV space.

9 Conclusion

The Ω -Symmetry Hypothesis survives its first empirical test. In a single dataset of 20 subjects, the two predicted symmetry classes appear in human EEG phase coherence dynamics with the Fisher z -corrected CV ratio of 2.06 ± 0.19 matching the predicted value of 2.0 to within 2.8%. The pattern replicates across independent subjects ($r = 0.956$, $p = 0.003$) and is stable across the train–test split (train: 2.09, test: 2.01). The systematic attenuation observed in raw PLV values is explained by bounded-domain compression and is addressed by the Fisher z -transform, though the choice of correction affects the precision of the match (Table 6).

Three genuine failures require acknowledgement: the original band-to-class assignments were wrong, the regeneration amplitude showed a reversed sign due to bounded-domain effects, and the rate–CV correlation had the wrong sign. These are not fatal to the framework, but they prevent any claim of complete validation. The first requires the principled derivation provided in Section 4; the second may yield to the same bounded-domain correction that addressed the CV ratio; the third requires refinement of the simple threshold model. The framework also involves modelling assumptions beyond its four core commitments: the exponential kernel (Section 1.2), the $\Omega = 1/\phi$ functional form (Section 2.1), the $CV = \Omega/2$ scaling (Section 2.3), and the $\Omega = \sigma^2$ identification (Section 4.4). These are flagged as testable components that could be revised independently of the core CRR structure.

The honest summary: CRR’s mathematical structure corresponds to measurable dynamics in real brains. The central quantitative prediction (CV ratio = 2.0) is supported after correction for the bounded observable, and the conjunction of two-class structure, class-level ordering, train–test replication, and rupture-consequence scaling is not easily explained by alternative accounts. However, these results rest on a single dataset with acknowledged measurement limitations, and cross-dataset replication at higher sampling rates is essential before strong conclusions can be drawn. The Ω -Symmetry Hypothesis remains a hypothesis with promising initial support.

Data and Code Availability. All analyses were conducted in Python using MNE-Python, SciPy, and NumPy. The PhysioNet EEGBCI dataset is publicly available. Analysis code is provided in Appendix A and at [https://github.com/\[hinrichs-username\]/crr-eeg-validation](https://github.com/[hinrichs-username]/crr-eeg-validation).

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Authors’ Contributions. A.S.: conceptualisation, methodology, formal analysis, software, writing (original draft). N.H.: methodology, validation, writing (review and editing). X.C.: methodology, writing (review and editing).

Competing Interests. The authors declare no competing interests.

Use of AI. AI-assisted tools (Claude, Anthropic) were used for literature review, code development, and manuscript preparation. All scientific claims, theoretical arguments, and interpretive judgements are the responsibility of the named authors.

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A Analysis Code

The complete Python analysis pipeline comprises two scripts, available as supplementary files and at the repository linked above. The primary script (`crr_eeg_analysis.py`) downloads the PhysioNet EEGBCI dataset, computes windowed PLV across six frequency bands for 20 subjects with a train–test split, detects coherence ruptures, and generates figures 1–8. The bounded-domain correction script (`crr_cv_investigation.py`) replicates the analysis under both 5-band and 6-band schemes, applies Fisher z and logit transforms, and generates figures 9–10. Key parameters:

- PLV window: 1.0 s, step: 0.05 s
- Rupture threshold: 8% of 2 s running maximum
- Minimum inter-rupture interval: 0.3 s
- Bandpass: 0.5–60 Hz (FIR), 60 Hz notch
- Random seed: not applicable (deterministic pipeline)
- Software: MNE-Python 1.11.0, SciPy 1.17.0, NumPy, matplotlib